

Mohamed Boghdaddy

Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with hands-on experience building production-ready web platforms and AI-powered systems using MERN/MEAN stacks and Python-based AI services. Strong background in applied machine learning, prompt engineering, and microservices architecture. Proven ability to design scalable APIs, integrate LLMs into real products, and deliver end-to-end solutions from system design to deployment.

CORE TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, C, C++, Dart

Frontend: React, Next.js, Angular, React Native, HTML, CSS, TailwindCSS

Backend and APIs: Node.js, Express.js, REST APIs, Microservices Architecture

AI and Machine Learning: Prompt Engineering (Phi-2, Gemini, GPT), NLP, Time-Series Forecasting (LSTM, XGBoost, ARIMA), Computer Vision (YOLOv8, ViViT, VideoMamba), Model Evaluation and Optimization

Databases: MongoDB, PostgreSQL, Firebase

DevOps and Tools: Git, Docker, Linux (Ubuntu), Agile/Scrum, LangChain, Vite

PROFESSIONAL EXPERIENCE

Full-Stack AI Engineer — PreGen

Oct 2025 – Present

- Designed and developed a full MERN-based platform integrated with a Python AI engine for automated content generation.

- Built scalable REST APIs, authentication flows, and database schemas to support AI-driven workflows.
- Implemented AI microservices for prompt-based generation, evaluation, and automation.
- Owned system architecture decisions, service integration, and deployment workflows.

Prompt Engineer (Freelance)

Jul 2024 – Present

- Designed and optimized multi-turn prompt pipelines using few-shot learning and prompt chaining.
- Built LLM-powered solutions using Phi-2 and Gemini for financial reasoning and information extraction.
- Improved response consistency, contextual memory, and accuracy through prompt design.
- Collaborated with developers to integrate LLM workflows into production APIs.

UAT Engineer — Geidea for Technologies

Aug 2025 – Nov 2025

- Conducted User Acceptance Testing (UAT) for financial and payment platforms.
- Validated business requirements against system behavior and functional gaps.
- Reported and tracked defect resolutions across Agile sprints.
- Coordinated with engineering and product teams to ensure release readiness.

Full-Stack MERN Intern — ATOS

Jul 2024 – Oct 2024

- Engineered a Document Management System using MERN and PERN stacks.
- Developed and optimized RESTful APIs for secure document handling.
- Improved frontend performance using modular React components.
- Applied clean architecture practices to improve maintainability.

Microservices Software Developer Intern — Relief and Emergency Foundation

May 2024 – Aug 2024

- Implemented microservices architecture for WordPress and MERN platforms.
- Designed modular backend services using Node.js.
- Applied best practices for service separation and API communication.

Machine Learning Intern — Code Alpha

Feb 2024 – May 2024

- Developed ML models including recommender systems and stock forecasting.
- Applied data preprocessing, feature engineering, and model evaluation.
- Built LSTM-based time-series models using Python.

Web Development Intern — Prodigy Info Tech

Jan 2024 – Feb 2024

- Built responsive landing pages and interactive web applications.
- Developed a stopwatch web app with precise timing logic.

SELECTED PROJECTS

- PreGen — AI-powered content generation platform (MERN + Python AI).
- HKIApply — AI job application platform with resume analysis and interview prep.
- Financial AI Advisor — Phi-2 & Gemini powered budgeting and investment assistant.
- Medical Video Classification — ViViT vs VideoMamba deep learning research.
- Swarm_Negotiation— Automated Swarm Negotiation system for best prices using PSO.
- ThreadCRM — ReactNative-based CRM platform.
- Foundit — Flutter mobile app for lost and found

EDUCATION

Bachelor of Science in Computer Science

ECU — Specialization: Artificial Intelligence

COURSES AND CONTINUOUS LEARNING

Udemy: Agile & Scrum Project Management; Reinforcement Learning in Python; Hyperparameter Optimization; Microservices with Node.js and React; AI & Meta-Heuristics; Next.js 15 & React; NodeJS Projects Bootcamp; Python for Finance.

Coursera: Meta – Introduction to Front-End Development; Google – Foundations of UX Design.

LANGUAGES

English (Fluent), Arabic (Native), Spanish (Intermediate – A2)